



UNIVERSITÀ DI PADOVA

Dipartimento
di Ingegneria
dell'Informazione

Audio and Speech Compression

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Audio in the Multimedia Ecosystem

Compression: audio vs. video

While video accounts for the largest share of network traffic, audio imposes the strictest constraints on the communication chain.

- ▶ **The Latency Constraint:** Human auditory perception is significantly more sensitive to temporal jitter and delay than the visual system. Real-time interaction in immersive environments requires ultra-low end-to-end latency ($< 20\text{ms}$).
- ▶ **Dynamic Range and Precision:** Audio signals span a dynamic range of up to 100 dB. This necessitates high-precision quantization and adaptive strategies to maintain a constant signal-to-noise ratio.
- ▶ **Bandwidth vs. Transparency:** The objective is to achieve perceptual transparency, where the compressed signal is indistinguishable from the original, at minimal bitrates for high-density network scenarios.

Methodological Framework

Synergies with Signal Processing Foundations

The compression of audio signals integrates and specializes the fundamental coding tools developed throughout the course:

- ▶ **Perceptual Thresholding:** Moving from general sound perception to specific **Psychoacoustic Masking Models** to optimize the distribution of quantization noise.
- ▶ Extending the principles of **Predictive Quantization** to the physical modeling of the acoustic source, specifically the human vocal tract.
- ▶ **Transform Coding:** Adapting frequency-domain compaction techniques (like the DCT) to time-continuous signals via the **MDCT** (Modified Discrete Cosine Transform).
- ▶ **Entropy Optimization:** Applying advanced **Arithmetic or Huffman coding** to reach the information-theoretic limit of the quantized representation.

Methodological Taxonomy

Source Modeling vs. Sink Modeling

Compression strategies are formally categorized based on the specific end of the communication chain they prioritize:

1. **Source-Based Coding (Parametric)**: Exploits the physical constraints of sound generation. Modeling the vocal tract (LPC) enables high-intelligibility speech coding at extremely low data rates.
2. **Sink-Based Coding (Perceptual)**: Exploits the physiological limitations of the human ear. This is the foundation for high-fidelity audio and music (e.g., AAC, MP3).
3. **Hybrid Architectures**: Contemporary standards, such as **Opus** or **EVS**, dynamically integrate both paradigms to handle mixed content (speech and music) with maximal efficiency.

The Emerging Frontier

Neural Codecs based on generative models are redefining the lower bounds of bitrate by synthesizing high-quality audio from sparse latent representations.

Outline

Speech and Audio Foundations

Physical Modeling of the Human Source

Linear Predictive Coding (LPC)

Quantization and Stability: The LSF Representation

Source-Based Codecs

Standardization: AMR and EVS

Perceptual Audio Coding

Psychoacoustic Fundamentals

Analysis and Windowing

MPEG Standards: MP3 and AAC

The Modern Era: Opus and Neural Codecs

Conclusion and Evaluation

Quality Evaluation Metrics

Temporal Dynamics: Speech vs. Music

Variability of the Stationarity Assumption

Predictive and transform coding rely on signal stationarity

Audio exhibits local stationarity but, the duration of this stability window varies drastically across signal types:

- ▶ **Speech:** Characterized by a quasi-stationarity of approximately 20 ms, dictated by the physical inertia of the human vocal folds and articulators.
- ▶ **General Audio/Music:** Highly heterogeneous.
 - ▶ *Tonal segments:* (e.g., a sustained violin note) can remain stationary for hundreds of milliseconds.
 - ▶ *Transients:* (e.g., a drum hit) change their statistical properties in less than 2-5 ms.

The Compromise

Modern codecs must switch window lengths dynamically to avoid temporal artifacts like pre-echo during transients while maintaining frequency resolution for tonal parts.

Modeling Paradigms

Source Modeling vs. Sink Modeling

The choice of compression technique depends on the nature of the signal's redundancy:

- ▶ **Source-Based Modeling (Speech):** Effective when the generator is known (the vocal tract). We use AR filters to model resonances. It is extremely efficient but fails for non-vocal sounds. Priority on *intelligibility* and low latency.
- ▶ **Sink-Based Modeling (General Audio):** Since music lacks a single physical model, we focus on the receiver: the human auditory system. Priority on *fidelity* and perceptual transparency.

Redundancy vs. Irrelevancy

Speech coding focuses on exploiting **statistical redundancy** (prediction), while audio coding focuses on removing **perceptual irrelevancy** (masking).

Amplitude Statistics across Audio Types

PDF Sparsity and Dynamic Range

The efficiency of optimal quantizers (like Lloyd-Max) depends on the shape of the signal's Probability Density Function (PDF):

- ▶ **Sparsity:** Both speech and many musical instruments exhibit Laplacian-like PDFs (sharp peak at zero), indicating the signal spends most time at low energy.
- ▶ **Dynamic Range:** Music requires 16-24 bit precision to handle the vast range between silence and peaks, whereas speech is often acceptable with 8-bit non-uniform quantization.

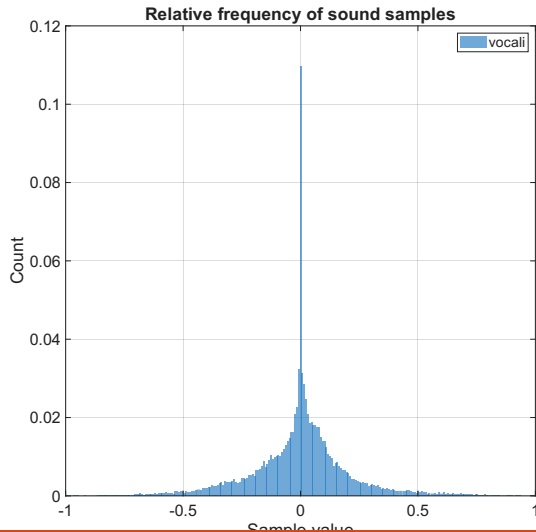
The following slides provide empirical evidence of these properties using real-world samples.

Empirical Analysis: Voiced Speech

Vowels - High Periodicity and Correlation

- ▶ **File:** `vocali.wav`
- ▶ **Analysis:** The histogram shows a multi-modal distribution reflecting the periodic nature of vowels.
- ▶ **Correlation:** The plot of consecutive samples (x_n vs x_{n-1}) shows a very tight diagonal alignment, indicating massive redundancy.

▶ Play Audio Sample

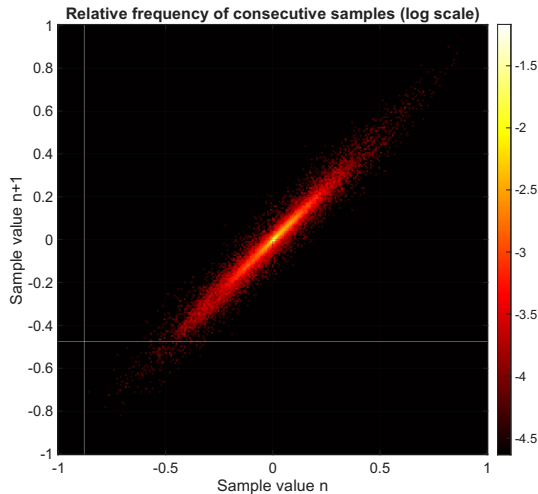


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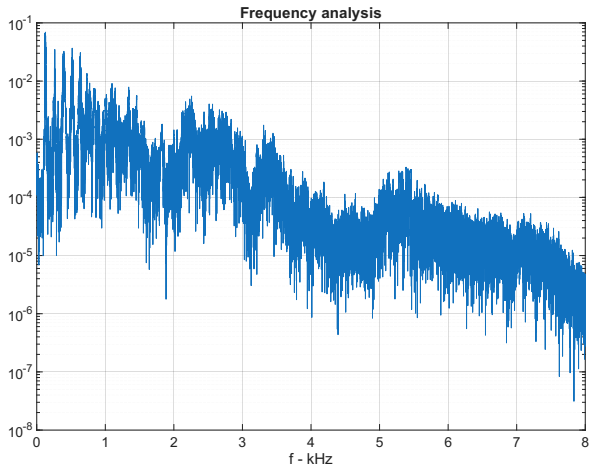


Spectral Analysis: Vowels

Formants and the Source-Filter Justification

Visualizing the Resonances:

- ▶ The Fourier analysis of vowel's sound clearly shows fundamental frequency and harmonics
- ▶ However, in order to take into account locality, we perform a *short-time Fourier Transform* over consecutive time windows

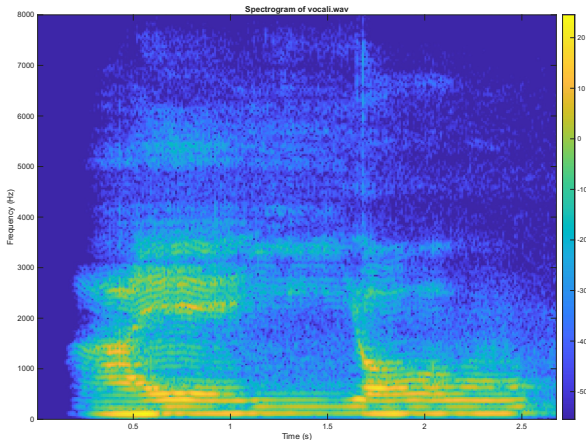


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- ▶ This produces the **spectrogram**, where we see:
- ▶ **Harmonic Structure:** The clear horizontal lines represent the fundamental frequency (pitch) and its harmonics.

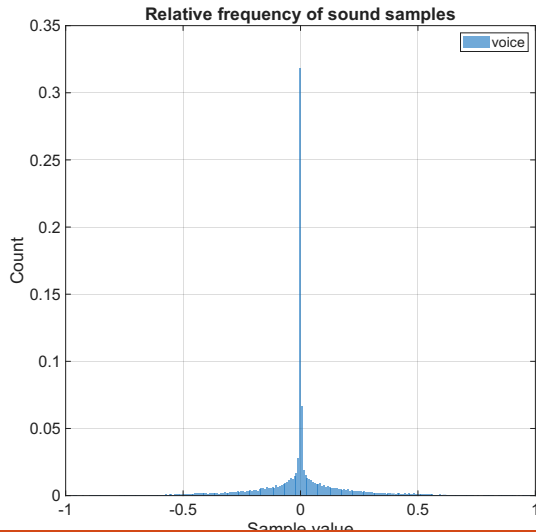


Empirical Analysis: General Speech

Mixed Phonemes and Laplacian Distribution

- ▶ **File:** `voice.wav`
- ▶ **Analysis:** A typical speech signal. The histogram exhibits the characteristic Laplacian "peaked" shape centered at zero.
- ▶ **Coding Insight:** The high density near zero confirms that a large portion of the signal can be represented with very few bits after prediction.

▶ Play Audio Sample

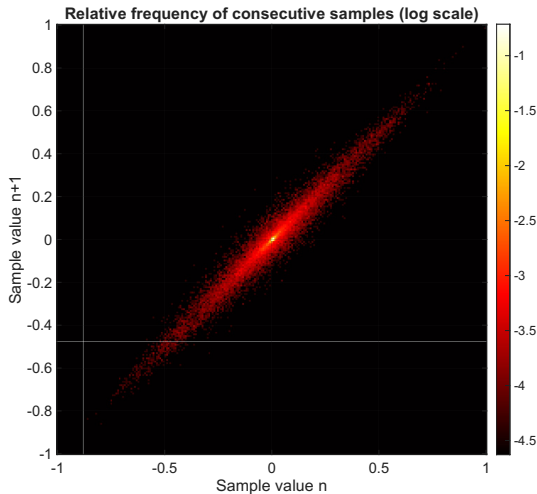


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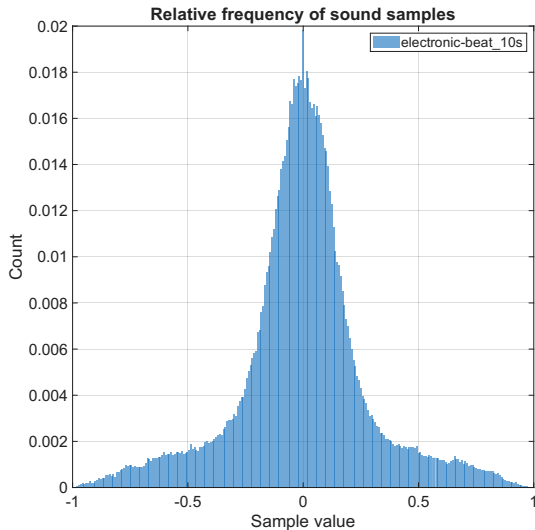


Empirical Analysis: Electronic Music

Beats and Transients

- ▶ **File:** `electronic-beat_10s.wav`
- ▶ **Analysis:** Percussive elements create more dispersion in the consecutive sample plot.
- ▶ **Challenge:** Sudden transients (the "beats") represent non-stationary events that break the prediction model.

▶ Play Audio Sample

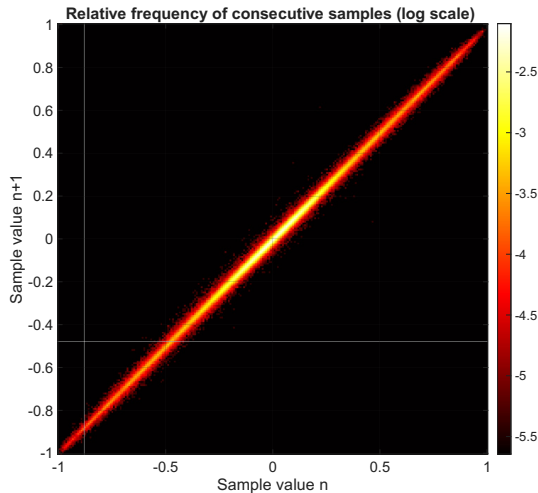


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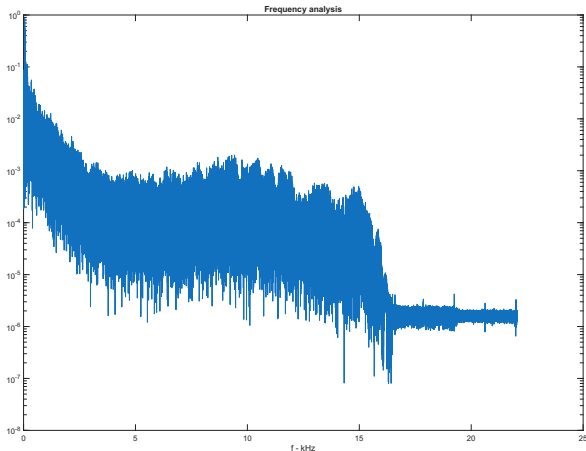
▶ Play Audio Sample



Spectral Analysis: Electronic Music

Absence of evident structures

- ▶ Frequency analysis of music signals does not show evident structures as those observed in speech
- ▶ The sharp fall of the amplitude spectrum at 16 kHz is due to an acquisition low-pass filter
- ▶ Using a spectrogram in this case would not improve our analysis

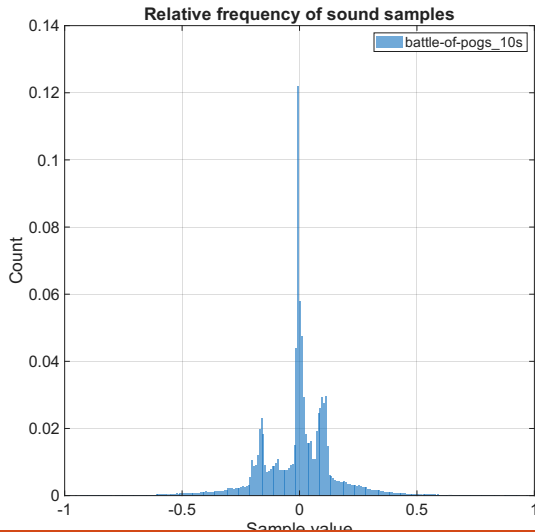


Empirical Analysis: Game Audio

Complex Soundscapes and Effects

- ▶ **File:** `battle-of-pogs_10s.wav`
- ▶ **Analysis:** The most "noise-like" distribution among the samples.
- ▶ **Complexity:** High-frequency components and non-harmonic textures make this signal harder to compress using parametric models.

▶ Play Audio Sample

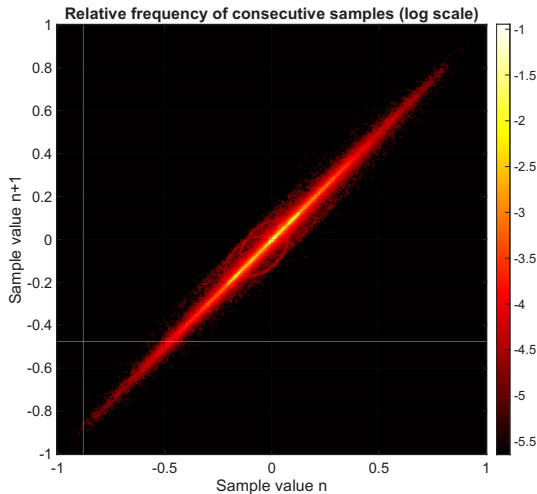


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Lessons from Empirical Data

Bridging Statistics to Coding Strategy

- ▶ **For Speech:** We can rely on source-filter models because the physical constraints of the vocal tract ensure good predictability.
- ▶ **For Music:** We must shift to perceptual models, as the time-domain signal is too unpredictable for pure source models.

The Prediction Gain (G_p)

By modeling the resonances we saw in the "voice" examples, we will now derive the **Linear Predictive Coding (LPC)** coefficients to maximize G_p .

The Human Speech Production System

From Anatomy to Signal Processing

Speech is generated through a highly structured mechanical process that can be mapped directly to a digital signal processing block diagram:

- ▶ **The Lungs (Power):** Provide the DC air pressure, acting as the system's power supply.
- ▶ **The Vocal Folds (Source):** The "sound generator." They vibrate periodically for voiced sounds or remain open for noise-like unvoiced sounds.
- ▶ **The Vocal Tract (Filter):** The pharynx, oral, and nasal cavities. These act as an acoustic tube, shaping the sound through resonances.

Voiced and Unvoiced Speech

The Two Modes of Acoustic Excitation

The human vocal system generates two fundamentally different types of acoustic excitation based on the state of the glottis:

- ▶ **Voiced Sounds** (e.g., vowels like /a/, /u/, or nasals like /m/):
 - ▶ **Mechanism**: The vocal folds are held together and vibrate periodically due to airflow
 - ▶ **Signal**: Quasi-periodic waveform with a clear **Pitch Period** (T_0).
 - ▶ **Spectrum**: Harmonic structure with energy concentrated at multiples of $F_0 = 1/T_0$.
- ▶ **Unvoiced Sounds** (e.g., fricatives like /s/, /f/, /sh/):
 - ▶ **Mechanism**: The vocal folds remain open (abducted); turbulence is created by air rushing through a narrow constriction.
 - ▶ **Signal**: Stochastic, noise-like waveform with no apparent periodicity.
 - ▶ **Spectrum**: Continuous and "flat" or "colored" (depending on the vocal tract shape), similar to white noise.

Impact on Coding

The encoder must distinguish between these two states: **voiced** frames require a pitch-period estimation (T_0), while **unvoiced** frames are modeled using a random noise generator.

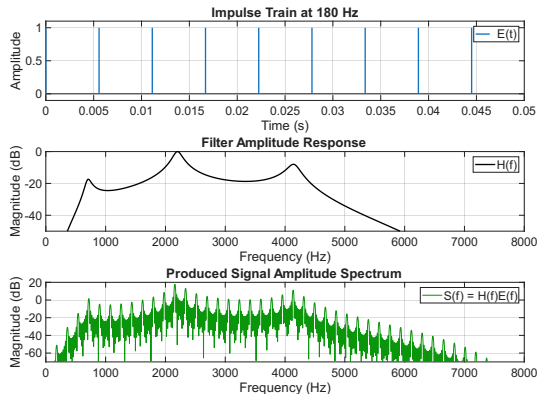
The Source-Filter Acoustic Model

Decoupling Excitation and Resonance

Fant's classic theory states that the speech signal s is the convolution of an excitation source e and a time-varying vocal tract filter h . In the frequency domain:

$$S(f) = E(f) \cdot H(f)$$

Instead of encoding the complex waveform s , we aim to transmit the filter parameters that represent the vocal tract and a simplified excitation model: impulse train for voiced, white noise for unvoiced.



Bridge to Linear Predictive Coding

- ▶ In the upcoming section, we will learn how to model the h filter by using predictive coding
- ▶ The model is based on a number P of discrete past samples
- ▶ If the prediction model is robust, the residual can be easily modeled either as noise or as a train impulse
- ▶ We work in the discrete domain: the input signal is $x(n)$, and we will use the Z-transform to describe the processing architecture
- ▶ The predictor is a discrete linear filter (turns out to be related to H)
- ▶ **LPC: Linear Predictive Coding**

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Principles of Linear Prediction

Modeling the Signal and the Residual

Following the Source-Filter model, we assume the current signal sample $x(n)$ can be estimated from a linear combination of its P past values:

$$\hat{x}(n) = - \sum_{i=1}^P a_i x(n-i)$$

The prediction error, or **residual signal** $y(n)$, is defined as:

$$y(n) = x(n) - \hat{x}(n) = \sum_{i=0}^P a_i x(n-i) \quad \text{with } a_0 = 1$$

The Whitening Filter

The residual $y(n)$ is the output of a FIR filter $A(z) = \sum_{i=0}^P a_i z^{-i}$ applied to $x(n)$. If the prediction is optimal, $y(n)$ is a white-noise-like signal or a periodic impulse train.

Estimation of Optimal Coefficients

Minimization of the Residual Variance

To find the coefficients $\{a_i\}$, we minimize the variance of the residual $\sigma_y^2 = E\{y^2(n)\}$. As seen in the Predictive Quantization module, setting the partial derivatives $\frac{\partial \sigma_y^2}{\partial a_i} = 0$ leads to the **Yule-Walker** equations:

$$\mathbf{R}_x \mathbf{a} = -\mathbf{r}_x$$

where:

- ▶ $\mathbf{R}_x$ is a symmetric **Toeplitz matrix** of autocorrelations.
- ▶ $\mathbf{r}_x$ is the autocorrelation vector
- ▶ $\mathbf{a} = [a_1, a_2, \dots, a_P]^T$ is the vector of LP coefficients.

Levinson-Durbin Recursion

The Toeplitz structure allows solving this system with $\mathcal{O}(P^2)$ complexity, making real-time estimation feasible even on low-power devices.

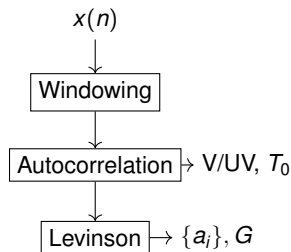
The LPC Analysis Process

From Waveform to Parametric Representation

The encoder segments the signal $x(n)$ into frames (approx. 20 ms) and extracts:

Main Parameters:

1. **LP Coefficients** $\{a_i\}$: define the spectral envelope (vocal tract).
2. **Gain** G : represents the signal energy (σ_y).
3. **V/UV Decision**: determines the source type.
4. **Pitch Period** T_0 : fundamental frequency (if voiced).



The Autocorrelation Estimation

The short-time autocorrelation $R_x(k)$ is the fundamental tool for extracting both source and filter parameters from a windowed frame $x_w(n)$.

- ▶ **Biased Estimator:** As seen in previous modules, we use:

$$R_x(k) = \sum_{n=0}^{N-1-k} x_w(n)x_w(n+k) \quad \text{for } k = 0, \dots, K$$

- ▶ **Filter Estimation (Lags $0 \dots P$):** These values form the $\mathbf{R}_x$ and $\mathbf{r}_x$
- ▶ **Source Analysis (Long Lags):** The same function is used for:
 - ▶ **Pitch Detection:** Finding the maximum peak at $k > P$.
 - ▶ **V/UV Decision:** Comparing the peak magnitude against $R_x(0)$.

Computational Efficiency

By computing $R_x(k)$ once, the encoder determines the spectral envelope, the signal energy, the periodicity (pitch), and the excitation type.

Characteristics of the Residual $y(n)$

What remains after spectral flattening?

The residual $y(n)$ represents the information that the linear filter cannot capture – the excitation source:

- ▶ **Voiced segments:** $y(n)$ shows sharp, periodic spikes (glottal pulses). The prediction gain is high because the signal is highly redundant.
- ▶ **Unvoiced segments:** $y(n)$ resembles zero-mean white noise. The linear model captures the "color" of the turbulence.

The compression efficiency

By transmitting the filter $A(z)$ and a simplified model of $y(n)$, we achieve high compression because $y(n)$ has a much lower variance and flatter spectrum than $x(n)$.

LPC Synthesis

Reconstructing the Signal via Inverse Filtering

The synthesis process is the exact mathematical inverse of the analysis. If the prediction error filter $A(z)$ is used to extract the residual $y(n)$ from the signal $x(n)$:

$$Y(z) = A(z)X(z) \implies X(z) = A^{-1}(z)Y(z)$$

At the decoder, we reconstruct the speech by applying the **inverse filter** $A^{-1}(z)$ to the excitation:

- ▶ **Excitation Source:** $y(n)$ is reconstructed as either a periodic impulse train (voiced) or white noise (unvoiced) based on the received parameters (Gain, Pitch, V/UV).
- ▶ **Time-Domain Implementation:** The synthesis is performed using a linear recurrence equation:

$$x(n) = y(n) - \sum_{i=1}^P a_i x(n-i)$$

Since $X(z) = A^{-1}(z)Y(z)$, the general shape of X is determined by $A^{-1}(z)$, since Y is either flat (white noise) or a Dirac's comb, see pag. 18

LPC modeling is more accurate near spectral peaks (formants) than in spectral valleys.

Notice that $A(z) = a_0 + a_1 z^{-1} + \dots + a_P z^{-P}$ is a FIR filter.

Therefore, its inverse is an all-pole, IIR filter.

The Quantization Problem

Why not quantize $\{a_i\}$ directly?

To transmit the vocal tract model $A(z)$, we need to quantize the filter coefficients $\{a_1, a_2, \dots, a_p\}$. However, direct scalar quantization of $\{a_i\}$ is extremely dangerous:

- ▶ **High Dynamic Range:** The coefficients a_i can vary wildly in magnitude, making uniform quantization highly inefficient.
- ▶ **Lack of Boundedness:** There is no simple relationship to guarantee that the quantized coefficients $\hat{a}_i$ will keep the roots of $A(z)$ inside the unit circle.

The Stability Constraint

If even one pole of the synthesis filter $1/A(z)$ moves outside the unit circle due to quantization noise, the decoder filter becomes unstable, resulting in massive audio distortion.

Line Spectrum Frequencies (LSF)

The Itakura Solution

Introduced by Fumitada Itakura (1975), LSFs map the LPC polynomial $A(z)$ into two new symmetric and anti-symmetric auxiliary polynomials:

$$P(z) = A(z) + z^{-(P+1)}A(z^{-1})$$

$$Q(z) = A(z) - z^{-(P+1)}A(z^{-1})$$

This decomposition creates a mathematical "miracle" for quantization.

Property 1: Roots on the Unit Circle

All roots of $P(z)$ and $Q(z)$ lie **exactly on the unit circle**. Therefore, they can be represented simply by their angles (frequencies) $\omega_i \in [0, \pi]$, without needing a magnitude component.

The Interlacing Property

Let $\omega_i^{(P)}$ be the roots of $P(z)$ and $\omega_i^{(Q)}$ be the roots of $Q(z)$.

Property 2: Strict Interlacing

The synthesis filter $1/A(z)$ is strictly stable **if and only if** the roots of $P(z)$ and $Q(z)$ interlace (alternate) along the unit circle:

$$0 < \omega_1^{(P)} < \omega_1^{(Q)} < \omega_2^{(P)} < \omega_2^{(Q)} < \dots < \pi$$

Why this is perfect for coding:

- ▶ We quantize the frequencies ω_i .
- ▶ If quantization error causes two frequencies to swap order, the decoder detects the anomaly instantly.
- ▶ The decoder simply re-sorts them to restore strict interlacing, guaranteeing a 100% stable filter.

From LSFs back to LPC Coefficients

Reconstructing the Filter at the Decoder

Once the decoder receives the quantized LSF angles $\{\omega_i\}$, it reconstructs the filter $A(z)$ to perform synthesis:

1. **Form the auxiliary polynomials:** The roots are placed on the unit circle at the received angles:

$$P(z) = (1 + z^{-1}) \prod_{i=1,3,\dots}^{P-1} (1 - 2 \cos(\omega_i) z^{-1} + z^{-2})$$

$$Q(z) = (1 - z^{-1}) \prod_{i=2,4,\dots}^P (1 - 2 \cos(\omega_i) z^{-1} + z^{-2})$$

2. **Reconstruct $A(z)$:** Based on the initial definitions, we simply average the two polynomials:

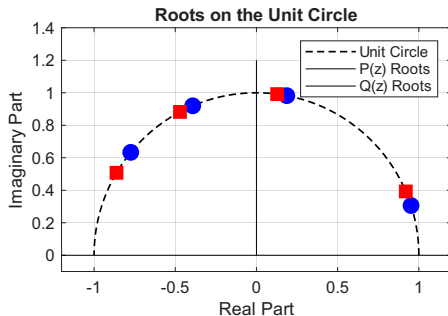
$$A(z) = \frac{P(z) + Q(z)}{2} \quad (1)$$

Visualizing LSFs and Formants

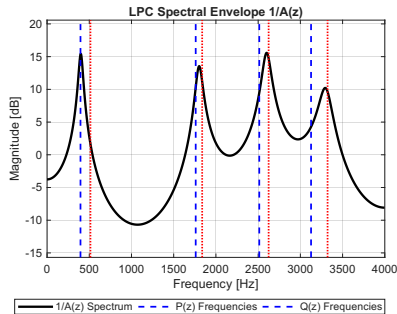
The LSF representation offers a direct visual link between the root positions and the spectral peaks (formants) of the vocal tract.

1. Z-Plane (Unit Circle):

- ▶ Blue roots: $P(z)$
- ▶ Red roots: $Q(z)$



2. Spectral Envelope: Resonance: When a P -root and a Q -root are very close together, the filter $A(z)$ exhibits a sharp spectral peak (a formant).



Efficient Coding: Vector Quantization (VQ)

Instead of quantizing each LSF ω_i independently, modern codecs use **Vector Quantization (VQ)**.

- ▶ **The Concept:** The entire set of P LSFs is treated as a single vector $\omega = [\omega_1, \dots, \omega_P]$.
- ▶ **The Codebook:** Encoder and decoder share a "dictionary" of representative LSF vectors, trained on human speech.
- ▶ **The Search:** The encoder finds the vector in the codebook that minimizes distortion and transmits only its **Index**.

Advantage

LSFs are highly correlated because the vocal tract can only take certain physical shapes. VQ exploits this to encode the whole 10th-order filter with just 20-30 bits per frame.

LPC-10 Bitrate Breakdown

The LPC-10 standard (Federal Standard 1015) operates at a constant bitrate of 2.4 kbps (54 bits per 22.5 ms frame). The bit allocation adapts to the source type:

Parameter	Voiced (Bits)	Unvoiced (Bits)
LPC Coefficients	41 (Order $P = 10$)	20 (Order $P = 4$)
Pitch Period (T_0)	7	0 (Reserved)
Energy (Gain G)	5	5
Synchronization	1	1
Error Protection	0	28
Total Bits / Frame	54	54

Observations

- ▶ **Voiced:** Maximum spectral detail ($P=10$) and pitch information.
- ▶ **Unvoiced:** Lower spectral detail ($P=4$) allows for significant error protection bits to improve robustness against channel noise.

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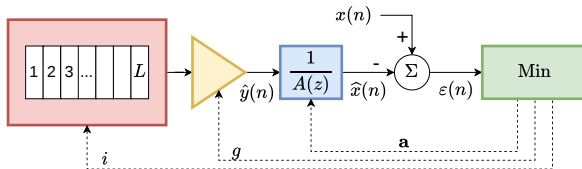
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The CELP Paradigm

From Open-Loop to Analysis-by-Synthesis

LPC-10 failed to achieve high quality because the excitation $y(n)$ was too simplistic. **CELP** (Code Excited Linear Prediction) introduces three innovations:

1. **Vector Excitation:** The residual $y(n)$ is no longer just pulses or noise, but a vector chosen from a **Codebook** (dictionary).
2. **Analysis-by-Synthesis (AbS):** The encoder acts as a "mini-decoder". It tries different excitation vectors i and different gains g , then synthesizes the speech, and compares it to the original.
3. **Closed-Loop Search:** The shape/gain of the vector that produces the minimum error are transmitted.



Perceptual Weighting Theory

Hiding Noise in the Formants

The previous scheme is a simplification made to introduce the main concepts of CELP encoders.

To improve the Quality of Experience, CELP does not minimize the physical error $\epsilon(n) = x(n) - \hat{x}(n)$. It minimizes a **Perceptually Weighted Error**.

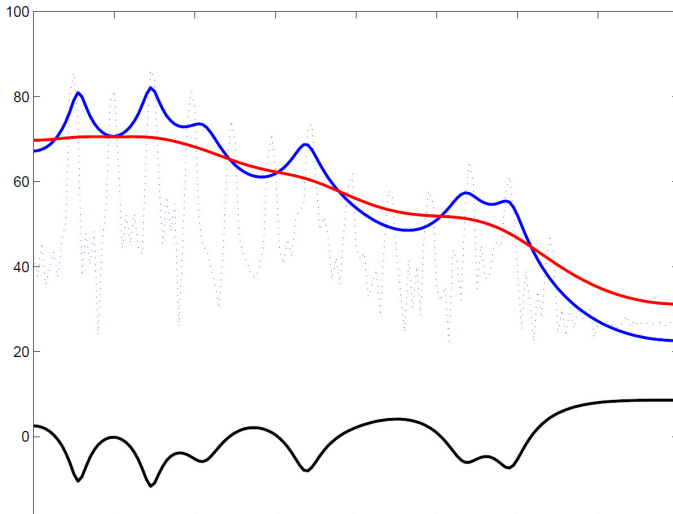
- ▶ **The Masking Principle:** The human ear is less sensitive to noise in frequency regions where the signal energy is high (the *formants*).
- ▶ **The Weighting Filter:** CELP uses a filter $W(z)$ to "shape" the error:

$$W(z) = \frac{A(z)}{A(z/\gamma)} \quad \text{with } 0 < \gamma < 1 \quad (2)$$

- ▶ **Pole Shifting:** By multiplying the filter coefficients by γ^j , we move the poles toward the origin, effectively **broadening the formants**.

Visualizing the Perceptual Mask

Spectral Shaping of the Quantization Noise



Blue: The original LPC spectral envelope $1/A(z)$.

Red: The broadened envelope $1/A(z/\gamma)$.

Black: $W(z)$
de-emphasizes the error near the formants, where the signal provides natural masking, and emphasizes it in the spectral valleys.

The CELP Excitation Model (1/2)

The Adaptive Codebook and Long-Term Prediction

The adaptive codebook models the **periodicity** (pitch) of voiced speech by "recycling" the previous excitation $y(n)$.

- ▶ **The Buffer:** A memory of the excitation $y(n)$ synthesized in previous subframes.
- ▶ **The Lag Q (Pitch Period):** The "adaptive" vector is extracted by going back Q samples in the buffer:

$$y_{\text{adapt}}(n) = y(n - Q)$$

- ▶ **The Gain $b = g_p$ (Pitch Gain):** Scales the past excitation to match the current energy.

Mathematical Interpretation

This acts as a **Long-Term Predictor (LTP)**. It effectively implements a filter $B(z) = g_p z^{-Q}$ that models the harmonic structure of the signal.

The CELP Excitation Model (2/2)

The Fixed Codebook and Multi-Shape Innovation

Since the past cannot predict everything (e.g., transitions or unvoiced noise), a **Fixed Codebook** provides the "innovation".

To achieve high quality without requiring an impossibly large dictionary, the innovation is constructed by combining multiple entries:

- ▶ **The Structure:** A static dictionary of shape vectors (e.g., algebraic pulse patterns).
- ▶ **Multi-Shape Combination:** Instead of choosing just one vector, the encoder selects **two shape vectors** (v_1 and v_2).
- ▶ **Independent Gains:** Each shape vector is scaled by its own gain factor (g_{c1} and g_{c2}).

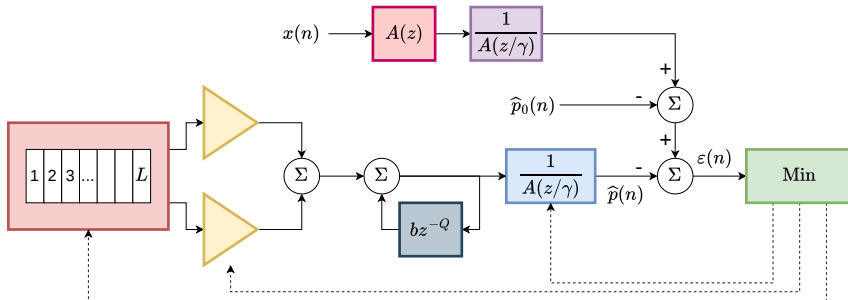
Total Excitation Construction

The final excitation $y(n)$ sent to the synthesis filter $1/A(z)$ is the sum of the adaptive past and the multi-shape innovation:

$$y(n) = \underbrace{g_p \cdot y_{adapt}(n)}_{\text{Periodic}} + \underbrace{g_{c1} \cdot v_1(n) + g_{c2} \cdot v_2(n)}_{\text{Fixed Innovation}} \quad (3)$$

Global CELP Architecture

Implementation in the ITU-T G.729 Standard



- **Feedback Loop:** The total excitation $y(n)$ is fed back into the adaptive codebook to be used as history for subsequent subframes.
- **Subframe Processing:** The excitation parameters (Q , v_1 , v_2 , g_p , g_{c1} , g_{c2}) are updated every 5 ms (subframe) to capture rapid transients, while the LPC filter $A(z)$ is updated every 10 ms (frame).

The Analysis-by-Synthesis Loop

Filter Memory and the Target Signal ($\hat{p}_0$)

In the G.729 diagram, before the error is minimized, a signal $\hat{p}_0(n)$ is subtracted from the perceptually weighted input. This is a crucial computational optimization.

- ▶ **Filter Components:** The output of the synthesis filter $\frac{1}{A(z/\gamma)}$ is the sum of two responses:
 1. **Zero-Input Response (ZIR):** The "ringing" due to the filter's past memory (states from previous subframes).
 2. **Zero-State Response (ZSR):** The response to the *new* excitation, assuming empty memory.
- ▶ **The Signal $\hat{p}_0(n)$:** This is exactly the ZIR. It is the output the filter would produce if the current codebook excitation were zero.

The Target Signal

By subtracting the past memory from the original weighted signal ($Target = x_{weighted} - \hat{p}_0$), the codebook search is tremendously simplified. The encoder now only needs to match the new codebook vectors (ZSR) to this memoryless Target Signal.

Bitrate Breakdown for G.729

8 kbps Allocation (80 bits per 10 ms frame)

Parameter	Update Rate	Bits
LPC Coefficients (LSF)	10 ms (Once/Frame)	18
Adaptive Codebook Index	5 ms (Twice/Frame)	14 (8 + 6)
Fixed Codebook Index	5 ms (Twice/Frame)	26 (13 + 13)
Codebook Gains	5 ms (Twice/Frame)	14 (7 + 7)
Parity/Sync	10 ms	8
Total		80 bits

Calculation

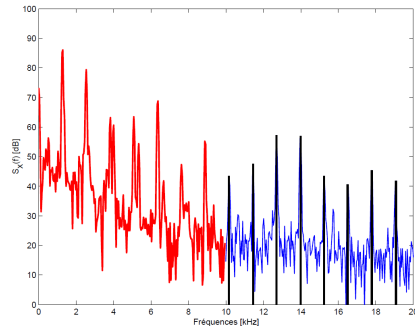
$$R = \frac{80 \text{ bits}}{0.010 \text{ s}} = 8000 \text{ bps}$$

The Wideband Revolution: AMR-WB

Standard 3GPP G.722.2 (HD Voice)

AMR-WB marked the transition from Narrowband (NB) to Wideband (WB) audio.

- ▶ **Extended Bandwidth:**
 - ▶ **50–200 Hz:** Provides "presence".
 - ▶ **3.4–7 kHz:** Improves fricatives and understandability.
- ▶ **ACELP Evolution:** Modified perceptual weighting and a significantly larger dictionary.
- ▶ **Adaptive Rates:** 9 bitrates (6.6 to 23.85 kbps).



Comparison of NB (dotted) vs WB (solid)
frequency response.

Enhanced Voice Services (EVS)

The Standard for VoLTE and 5G

EVS is the successor to AMR-WB, designed for high-fidelity audio over IP-based packet networks.

- ▶ **Bandwidth Versatility:** From Narrowband (NB) to **Full-band (FB, 20 kHz)**.
- ▶ **Hybrid Coding:** Integrates ACELP (speech) and MDCT (music) modes.
- ▶ **Bitrate Range:** Highly flexible (5.9 kbps to 128 kbps).

Backwards Compatibility

EVS includes an **AMR-WB IO** (Input/Output) mode to remain compatible with older devices while improving the transport robustness.

Solving IP Network Challenges

EVS is optimized for Packet-Switched (PS) networks where latency and loss are common:

- ▶ **Jitter Buffer Management:** New algorithms to smooth out the arrival of IP packets.
- ▶ **Frame Erasure Concealment (FEC):** Advanced techniques to hide packet loss by predicting speech from past parameters.
- ▶ **Discontinuous Transmission (DTX):** Optimized Silence Insertion Descriptors (SID) and comfort noise.

Mobile Standards Comparison

The Path from 2G to 5G

Standard	Network	Quality	Bandwidth
GSM-FR	2G	Telephony	300–3400 Hz
AMR-NB	3G	Telephony	300–3400 Hz
AMR-WB	4G/LTE	HD Voice	50–7000 Hz
EVS	4G/5G	Hi-Fi	20–20000 Hz

The Convergence

EVS represents the final merge of speech-specific (ACELP) and general audio (MDCT) technologies, see next sections

Outline

Speech and Audio Foundations

Physical Modeling of the Human Source

Linear Predictive Coding (LPC)

Quantization and Stability: The LSF Representation

Source-Based Codecs

Standardization: AMR and EVS

Perceptual Audio Coding

Psychoacoustic Fundamentals

Analysis and Windowing

MPEG Standards: MP3 and AAC

The Modern Era: Opus and Neural Codecs

Conclusion and Evaluation

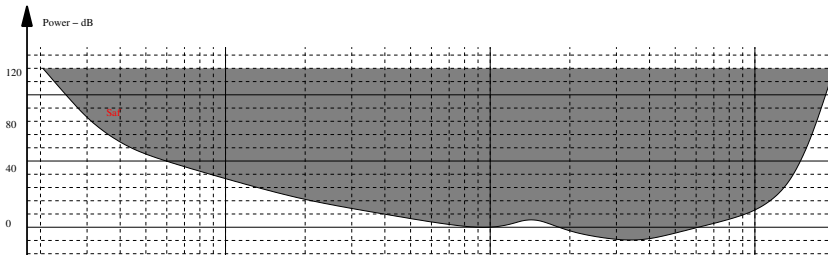
Quality Evaluation Metrics

Absolute Threshold of Hearing

Minimum Audible Pressure

The human ear does not have a flat frequency response. The power required for a sound to be audible, $S_a(f)$, varies significantly with frequency.

- ▶ **Audible Range:** Typically between 20 Hz and 20 kHz.
- ▶ **Maximum Sensitivity:** The threshold is lowest (most sensitive) between 1 kHz and 4 kHz, which corresponds to the core frequencies of human speech.
- ▶ **Coding Implication:** Sounds with power $\sigma^2 < S_a(f)$ are not heard and can be discarded by the encoder without any loss in subjective quality.

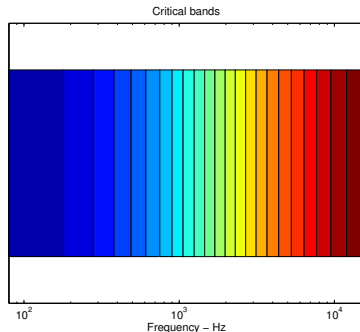


Critical Bands (BC)

The Bark Scale

The cochlea acts as a bank of overlapping bandpass filters. These filters define the **Critical Bands**.

- ▶ **Definition:** If multiple sinusoids fall within the same critical band, the ear perceives their total power rather than individual components.
- ▶ **Variable Width:** The bandwidth of these filters increases with frequency (narrower at low frequencies, wider at high frequencies).
- ▶ **Standardization:** AAC uses 51 groups to approximate these critical bands.

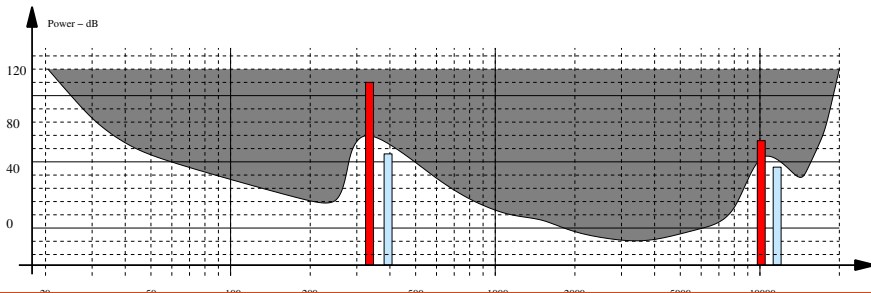


Frequency Domain Masking

Masking Curves $S_m(f_0, \sigma^2, f)$

A strong "masker" sound reduces the sensitivity to a second "masked" sound at nearby frequencies.

- ▶ **Masking Threshold:** For a given frequency f_0 and power σ^2 , the threshold $\Phi(f)$ increases, hiding noise below it.
- ▶ **Asymmetry:** Masking is more effective for frequencies higher than the masker's frequency ($f > f_0$).
- ▶ **Coding Gain:** We allow the quantization noise $S_Q(f)$ to increase as long as it remains below the masking curve: $S_Q(f) < \Phi(f)$.

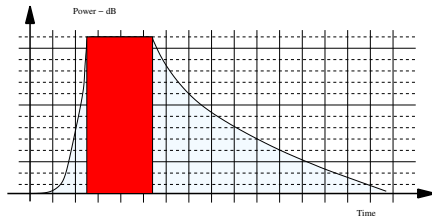


Temporal Domain Masking

Pre-masking and Post-masking

The masking effect also occurs in time, before and after a loud sound.

- ▶ **Pre-masking:** Lasts about 2–5 ms. The ear is "blinded" just before the onset of a loud sound.
- ▶ **Post-masking:** Much longer effect, lasting 100–200 ms. Sensitivity recovers slowly after a sound stops.
- ▶ **Windowing:** Temporal masking justifies the use of specific window shapes to avoid "pre-echo" artifacts in transients.



The MPEG Legacy

The most successful perceptual coders are part of the ISO/IEC MPEG (Moving Picture Experts Group) family:

- ▶ **MP3**: Formally known as **MPEG-1 Audio Layer 3**. It uses a hybrid filterbank (polyphase + MDCT).
- ▶ **AAC**: Stands for **Advanced Audio Coding**. It is defined in the **MPEG-2** and **MPEG-4** standards.

Design Objective: Achieve "transparent" quality (indistinguishable from the original) by removing irrelevant information based on the psychoacoustic model.

Transform Coding Principles

Windowing and Overlap

Perceptual coders operate on discrete blocks of signal called windows.

- ▶ **Overlapping:** To avoid blocking artifacts, windows overlap in time.
- ▶ **Frame vs. Buffer:**
 - ▶ **MP3:** Uses a buffer of $N = 512$ samples, coding $M = 32$ new samples.
 - ▶ **AAC:** Uses a larger buffer of $N = 2048$ samples, coding $M = 1024$ new samples per frame.
- ▶ **Window Switching:** Modern encoders switch between "Long" windows (for steady state) and "Short" windows (for transients) to manage temporal masking.

Modified Discrete Cosine Transform (M-DCT)

The Engine of Modern Audio Coding

The M-DCT is a block-based transform used to map the signal from the time domain to the frequency domain.

- ▶ **Overlap:** It uses overlapping windows (typically 50%) to eliminate blocking artifacts at the frame boundaries.
- ▶ **TDAC:** *Time Domain Aliasing Cancellation*. The mathematical structure ensures that the aliasing introduced in one block is cancelled by the next during reconstruction.
- ▶ **Critical Sampling:** This is a key efficiency property.
 - ▶ Despite the 50% overlap (where we process $N = 2M$ samples), the transform produces only M spectral coefficients.
 - ▶ **Result:** The number of frequency coefficients equals the number of *new* time samples, meaning there is **no redundancy** or data expansion.

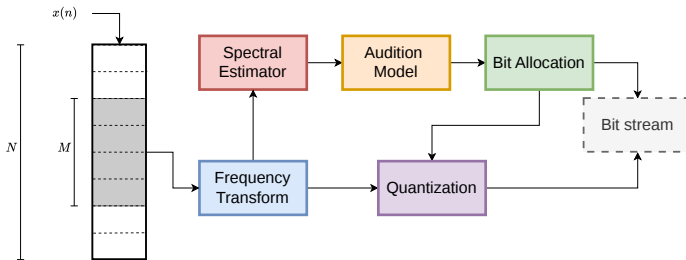
Generic Perceptual Coder Architecture

The Analysis-by-Synthesis Loop in Frequency

The encoder and decoder are asymmetric: most of the complexity lies in the encoder's bit allocation logic.

Encoder Tasks:

1. Time-Frequency mapping.
2. Psychoacoustic analysis.
3. Dynamic bit allocation.
4. Quantization and Huffman coding.



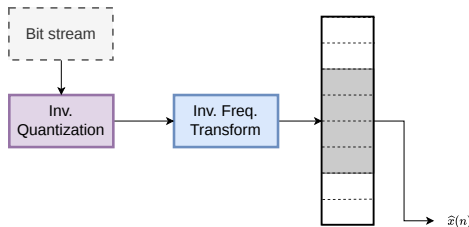
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MPEG-1 Layer 3 (MP3)

Signal-to-Mask Ratio (SMR)

The core efficiency of MP3 comes from its ability to measure the "perceptual headroom" in each subband to guide bit allocation.

Definition: Signal-to-Mask Ratio (SMR)

For a given frequency band k , the SMR is the ratio between the signal power S_k and the masking threshold Φ_k :

$$SMR_k[dB] = S_k[dB] - \Phi_k[dB]$$

- ▶ **Role:** SMR tells the encoder how much noise can be "injected" before it becomes audible.
- ▶ **Allocation Goal:** Assign enough bits to the subband so that the Signal-to-Noise Ratio (SNR_k) exceeds the SMR_k .
- ▶ **Transparency Condition:** When $SNR_k > SMR_k$, the quantization noise S_Q remains below the threshold Φ , becoming **inaudible**.

The Psychoacoustic Model: Calculating $\Phi(f)$

To find the global masking threshold $\Phi(f)$, the encoder follows a standardized psychoacoustic procedure:

1. **Spectral Analysis:** A high-resolution FFT is performed to identify the energy distribution.
2. **Identification of Maskers:** The model distinguishes between **Tonal** (sinusoids) and **Noise-like** components, as they have different masking properties.
3. **Individual Masking Curves:** For each masker, a "spreading function" (the masking tent) is calculated based on its intensity and frequency.
4. **Summation:** All individual curves are combined with the **Absolute Threshold of Hearing (ATH)** to form the **Global Masking Threshold $\Phi(f)$** .

MPEG-1 MP3 vs. MPEG-2/4 AAC

Evolution of Perceptual Audio Coding

AAC was designed to be the successor to MP3, fixing its structural limitations and providing better quality at lower bitrates.

Feature	MP3	AAC
Transform	Hybrid (Polyphase + MDCT)	Pure MDCT
Frequency Bands	32 uniform subbands	51 Critical Bands (SFB)
Resolution	Fixed / Limited	Flexible (up to 1024 bins)
Quantization	Simple / Linear	Non-linear (Power Law $X^{3/4}$)
Huffman Coding	Static tables	Dynamic (12 different tables)
Typical Bitrate	Transparent @ 192 kbps	Transparent @ 128 kbps

The scale factor Logic in AAC

Unlike MP3's rigid bands, AAC uses the Scale Factor (g) to surgically place the noise floor for each Scale Factor Band (SFB), ensuring that quantization noise S_Q is always hidden under the masking threshold Φ .

Spectral Band Replication (SBR)

Modern standards like **HE-AAC** (High Efficiency) and **USAC** go beyond simple quantization.

- ▶ **HE-AAC**: Combines AAC with SBR (to replicate high frequencies) and Parametric Stereo.
- ▶ **USAC**: A hybrid codec that switches between AMR-WB (for speech) and HE-AAC (for music) to handle mixed content seamlessly.

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Opus: The Hybrid King

Blending SILK and CELT

Opus is designed for the modern web, capable of handling everything from low-bitrate VoIP to high-fidelity music with very low latency.

- ▶ **SILK Layer:** Derived from Skype. Based on **LPC** (Linear Prediction), optimized for speech (8–12 kbps).
- ▶ **CELT Layer:** Derived from Xiph.org. Based on **MDCT**, optimized for audio and music (high bitrates).
- ▶ **The Hybrid Mode:** Between 16 and 32 kbps, Opus uses both layers simultaneously (SILK for the low frequencies, CELT for the highs).

Key Feature: Seamless Adaptation

Opus can change its bitrate, audio bandwidth (NB to FB), and coding mode on-the-fly without any audio glitches or renegotiation.

Opus: Low Latency for the Internet

Why it replaced MP3 and AAC for Real-Time

- ▶ **Algorithmic Delay:** While MP3/AAC often require 100+ ms, Opus can work with frames as small as **5 ms**.
- ▶ **VoIP Superiority:** This makes it the standard for WebRTC, Discord, and WhatsApp.
- ▶ **Resilience:** Includes advanced In-band Forward Error Correction (FEC) to survive packet loss on unstable Wi-Fi/4G connections.

The AI Frontier: Neural Codecs

From Quantization to Generation

Traditional codecs try to match the *waveform*. Neural codecs (like Google's **Lyra** or Meta's **EnCodec**) try to match the *perception* using Generative AI.

- ▶ **Encoder (Feature Extraction)**: A neural network compresses audio into a very small "latent space" (as low as 3 kbps).
- ▶ **Decoder (Generative Synthesis)**: Instead of simple inverse transforms, a **GAN** (Generative Adversarial Network) or a **Diffusion Model** "re-synthesizes" the voice.
- ▶ **Result**: Intelligible and natural-sounding speech at bitrates where CELP would sound like broken noise.

The Paradigm Shift

We no longer send the samples; we send "instructions" for the AI to reconstruct a plausible version of the original signal.

The Evolution of Spatial Audio

Channels, Objects, and Scenes

Modern immersive audio (VR/AR and Cinema) uses three different approaches:

- ▶ **Channel-based**: Fixed positions (5.1, 7.1). The mix is baked for a specific speaker layout.
- ▶ **Object-based (MPEG-H / Dolby Atmos)**: Sound sources (e.g., a "helicopter") are sent as mono signals + **XYZ metadata**. The receiver renders the sound based on the user's actual speakers.
- ▶ **Scene-based (Ambisonics)**: Captures the entire sound field sphere. Ideal for 360 videos and Head Tracking.

Interactivity and Personalization

The MPEG-H standard (used in ATSC 3.0 and DVB) introduces User Interactivity:

- ▶ **Object Control:** The user can manually increase the volume of the "Dialogue Object" or the "Stadium Commentary" while keeping the background noise at the same level.
- ▶ **Binaural Rendering:** Downmixes 3D audio for standard headphones, simulating a 22.2 speaker setup using HRTF (Head Related Transfer Functions).

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The Gold Standard: MOS and MUSHRA

The only true judge of audio quality is the human ear. Standardized tests include:

- ▶ **MOS (Mean Opinion Score):** Listeners rate clips from 1 (Bad) to 5 (Excellent). A score of 4.0 or higher is usually considered "toll quality".
- ▶ **MUSHRA (ITU-R BS.1534):** Used for intermediate quality. Listeners compare multiple samples against a "Hidden Reference" (original) and "Anchors" (low-pass filtered versions).
- ▶ **Transparency Test:** Used for high-end codecs (AAC). The goal is for the listener to be unable to distinguish the original from the compressed version in a double-blind "ABX" test.

Why SNR is Not Effective for Lossy Audio

Perceptual Noise vs. Mathematical Noise

In traditional DSP, we use Signal-to-Noise Ratio (SNR). In audio coding, this fails because:

- ▶ **Masking**: A codec might introduce 20 dB of noise that is perfectly masked by the signal (inaudible). The SNR would be low, but the quality perfect.
- ▶ **Phase Shifting**: Perceptual codecs often change the phase of the signal. The waveform looks different (high mathematical error), but it sounds identical.

The Shift

We need **Perceptual Metrics** that model the human auditory system to estimate the MOS automatically.

Objective Metrics: POLQA and ViSQOL

Replacing the Human Listener

Modern networks use algorithms to predict the MOS score in real-time.

- ▶ **POLQA (ITU-T P.863)**: The industry standard for 4G/5G. It compares the original and degraded signal using a psychoacoustic model (Bark scale, loudness mapping).
- ▶ **ViSQOL (by Google)**: A newer metric based on **spectrogram similarity**. It is particularly robust for VoIP and "generative" neural codecs where the time-alignment might be imperfect.

The Audio Coding Landscape

A Comparative Overview

Codec	Paradigm	Bitrate	Latency	Primary Use Case
AMR-WB	Source (CELP)	6-24 kbps	20-30 ms	Legacy Mobile Voice (3G/4G)
AAC	Receiver (MDCT)	64-128 kbps	>100 ms	Streaming and Storage
Opus	Hybrid	6-128 kbps	5-20 ms	WebRTC / Real-time Internet
EVS	Hybrid	5-128 kbps	32 ms	5G VoNR / VoLTE
Lyra/EnCodec	Neural	3-6 kbps	Variable	Ultra-low bandwidth / Fallback

Future Trends in Audio Coding

The Shift towards AI-Driven and Hybrid Systems

- ▶ **The Neural "Safety Net":** Today, AI codecs act as a fallback. If the network bitrate drops below 6 kbps, the system switches from waveform coding to generative reconstruction.
- ▶ **Hybridization:** Future standards will likely merge **MDCT** (for transparency in critical bands) with **Neural layers** (to "imagine" high-frequency textures or background noise).
- ▶ **Computation vs. Bandwidth:** A paradigm shift where we use more local battery/CPU power (NPUs) to save precious network bandwidth.
- ▶ **The "Hallucination" Challenge:** Balancing generative realism with mathematical transparency. In professional/legal fields, faithful reconstruction remains superior to AI-generated "plausible" audio.